PELSUNG TECH WING · POST-IMMERSION PROGRAMME · SELECTION

One-Day Build Challenge: SA Wellbeing Platform Take-home assignment · Candidate brief · Issued 4 November 2026

The challenge in brief

In this one-day take-home assignment, you will take a real problem the Tech Wing is about to work on and move it from idea to a working prototype, a short write-up and a 5-minute recorded pitch, working from home. The problem is helping school counsellors act on the Situational Analysis (SA) wellbeing survey.

You do not need a coding degree. **AI coding tools are allowed and expected.** We want to see how you think, build, explain and decide under a deadline, because that is what every week of the programme looks like.

The problem

Every year, students in many Bhutanese schools fill in the **Situational Analysis**, a short survey about how they are doing at home, at school and with friends. Only the school counsellor sees the answers and uses them to reach students who need help early.

Today each school handles the results on its own. A counsellor has to work out who needs help first, keep track of whom they have already met, and send reports upward. The Career Education & Counselling Division (CECD) of MoESD wants one platform that can:

- run the survey
- flag students who may be at risk
- track every follow-up
- show supervisors patterns across schools **without showing any student's identity**

Your job is to build the first small piece of that platform and convince us it is the right piece.

Privacy rule: this is children's mental-health data. In the challenge you use only the synthetic dataset we provide. Treat it as if it were real.

Your four deliverables

Each deliverable tests one skill. All four are due by the end of the day.

#	Deliverable	What it contains	Format	Skill tested
1	**Idea sheet**	Who the users are; their top 3 pain points; at least 5 ideas for a first feature; the one you pick and why; one thing in this brief you would challenge	1 page (PDF or photo of paper)	Brainstorming and ideation
2	**Working prototype**	A "counsellor flag board" built on the dataset (see acceptance criteria below)	Public GitHub repo that runs with one command, plus a live link if deployed	Coding
3	**README**	How to run it; what works and what does not; the rules and assumptions you used; how you used AI tools (one prompt that worked, one that failed); what you would build next	Markdown in the repo, max 1 page	Documentation
4	**Pitch video + update email**	A 5-minute screen-recorded demo and pitch (a phone recording is fine; your face is optional). Also a 150-word email to the CECD Director explaining what you built, in plain language with no jargon	Video link + text	Communication

You will also post a **progress update by 15:00** in the question channel, in 3 lines: what is done, what is next, what is blocking you. Shortlisted candidates are then invited to a **15-minute interview**, where you walk us through your code and make a small live change.

Prototype acceptance criteria

These use the same format as real Tech Wing problem statements. Finish the **Must** items before anything else. A smaller prototype that works beats a bigger one that breaks.

ID	Criterion	Priority
AC-01	The app loads the provided CSV and shows every student by ID, school and class (no names exist)	**Must**
AC-02	Each student gets a flag level (Urgent, High, Review or None) using the baseline rules, and the **reason** is shown next to the flag	**Must**
AC-03	Urgent students always appear at the top and are visually unmistakable	**Must**
AC-04	A counsellor can set each flagged student's follow-up status to Not started, In progress or Closed	**Must**
AC-05	A separate **supervisor view** shows flag counts by school and class, with no student IDs, and any group with fewer than 5 students shown as "<5"	**Must**
AC-06	Filter or search the counsellor list by school, class or flag level	**Should**
AC-07	Follow-up statuses survive a page refresh	**Should**
AC-08	You propose and implement one extra rule or improvement, and explain why in the README	**Should**
AC-09	Deployed to a public link, a chart on the supervisor view, Dzongkha labels, or a mobile-friendly layout	Could

The dataset and baseline rules

You receive **SA_sample_dataset_SYNTHETIC.csv**: 72 fictional students across 4 fictional schools, one row each. It contains no names or real data. Like real survey data, it is not perfectly clean.

Column	Values
student_id, school, class	e.g. S-001, School A, VI
q1_sleep_well	1 (rarely) to 4 (always)
q2_safe_at_home	1 (never) to 4 (always)
q3_bullied	Never, Sometimes, Often
q4_trusted_adult	Yes, No ("Is there an adult you can talk to?")
q5_sad_most_days	Yes, No
q6_thoughts_of_self_harm	Yes, No, Prefer not to say
q7_wants_counsellor	Yes, No
comment	Optional free text

Baseline flagging rules. Apply them in order; the first rule a student matches sets their flag.

1. **Urgent:** q6 = Yes
2. **High:** q3 = Often, or q2 = 1
3. **Review:** any two of (q1 ≤ 2, q5 = Yes, q4 = No), or q7 = Yes
4. **None:** everything else

These rules are a starting point, not the final word. If you think a rule is missing or wrong, say so in your idea sheet or README. Spotting a gap counts in your favour.

Rules, what you need and how to submit

Rules

- Work individually. You may use the internet, documentation and any AI tool. Declare every tool you used in the README.
- Use only the synthetic dataset. Never put real student information into any tool.
- You must be able to explain and change your own code. The shortlist interview includes a small live change; if you cannot explain your own submission, it is not counted.
- Ask questions in the channel between 08:00 and 20:00. Answers are shared with everyone. Good questions count in your favour.

How to submit: use the submission form linked in the release email. Send your repo link, your video link (unlisted YouTube, Loom or Google Drive with view access), your idea sheet and your email text. Name your files <your-name>-<deliverable> .

What happens next: results are shared before orientation on 16 November, 12 noon